Mphele Moswane

BACKEND SOFTWARE ENGINEER

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GitHub · LinkedIn · Portfolio

PROFILE

Final-year software engineering student focused on backend systems, scalable APIs and accessible technology. Three-time hackathon winner, including first place at the Geekulcha Annual Hackathon 2026, the ITWeb Security Summit and the Mjanga AI Hackathon. Builds reliable systems across Java, Go, Python and TypeScript. CompTIA A+ certified.

TECHNICAL SKILLS

Languages: Java, Python, Go, TypeScript, JavaScript, Bash
Frameworks: Spring Boot, FastAPI, Node.js
Data: PostgreSQL, MySQL, SQLite, MongoDB
Engineering: REST APIs, event-driven architecture, TDD, CI/CD, Docker, Linux, ML integration

SELECTED PROJECTS

Limi - Multilingual and SASL Accessibility Engine **1st Place, Geekulcha Annual Hackathon 2026**

OCR · native speech · semantic structure recovery · offline-first learning

Transforms textbooks, photographed notes, university PDFs and YouTube videos into accessible learning formats across 12 South African languages and South African Sign Language.

- Worked with a small team to build a universal ingestion flow for typed documents, photographed notes, PDFs and video content.
- Combined OCR with semantic structure recovery so headings, sections and reading order survive conversion instead of becoming unstructured text.
- Delivered synchronized native speech and a 3D SASL experience so learners can read, hear or see the same material signed in their language.
- Shaped offline-first learner modes and API-driven flows for students and institutions working with limited connectivity.

Doria - AI Supply Chain Security Tool **1st Place, ITWeb Security Summit 2026**

Python · TypeScript / Node.js · XGBoost | GitHub | Demo

A zero-setup CLI that scans npm packages before installation to catch malicious, typosquatted and AI-hallucinated dependencies without executing package code.

- Built a local forward proxy that automatically intercepts every package install, keeping protection inside the developer's normal terminal workflow.
- Trained an XGBoost model that scores risk from package metadata, install behaviour and name similarity to detect typosquatting attempts.
- Cross-referenced registry data and added package hashing to flag suspicious releases or malicious changes after an initially trusted version.

AfyaAI - Multilingual Medical Assistant **1st Place, Mjanga AI Hackathon 2026**

Python · FastAPI · Llama 3.1 · Google Cloud TTS | GitHub

A voice-first assistant that provides multilingual guidance during medical scares, calms users with actionable next steps and routes severe cases to nearby clinics.

- Built an asynchronous FastAPI backend that handles translation, text-to-speech and AI inference concurrently so multiple users can receive timely guidance.
- Added severity-parsing guardrails and geolocation routing that escalate only genuinely urgent cases while keeping the initial experience calm and accessible.
- Implemented a fault-tolerant speech pipeline with language fallbacks and synchronized visemes for the frontend's speaking 3D guide.

EDUCATION AND EXPERIENCE

WeThinkCode_ | Software Development Diploma Sep 2025 - Dec 2026

- Completed peer-reviewed projects covering Python fundamentals, REST and OAuth integrations, and Java OOP with client-server sockets, multithreading, Maven, JUnit and GitHub Actions.
- Worked in agile teams on brownfield codebases using TDD, refactoring, unit, acceptance and integration testing, SQL persistence, design patterns, CI/CD and Docker.

Peer Tutor | WeThinkCode_ Oct 2025 - Aug 2026

- Facilitated Code Clinics for first-year students, diagnosing bugs, explaining core programming concepts and guiding students toward test-driven solutions without taking over their work.
- Used code review and constructive peer feedback to strengthen students' problem-solving, code comprehension and technical communication.

CERTIFICATION AND AWARDS

CompTIA A+	Certification
Geekulcha Annual Hackathon 2026	1st Place - Limi
ITWeb Security Summit Hackathon 2026	1st Place - Doria
Mjanga AI Hackathon 2026	1st Place Africa-wide - AfyaAI